Yixiao Wang

Robot Learning | Model Architectures | Data-Centric Learning | Generalization | Generative Models

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PhD researcher at UC Berkeley studying efficient and generalizable robot learning, with work on model architectures, representation learning, data-centric policy learning, and the theory and acceleration of generative models.

Education

University of California, Berkeley Sep 2022 – May 2027 (expected)
PhD in Mechanical Engineering | Advisor: Prof. Masayoshi Tomizuka

Northwestern University Sep 2019 – Jun 2021
M.S. in Mechanical Engineering

Shanghai Jiao Tong University Sep 2014 – Mar 2021
B.S. in Mechanical Engineering, 2018; M.S. in Mechanical Engineering, 2021

Industry Research Experience

Amazon | Applied Scientist Intern May 2026 – Aug 2026

Pretrained reusable action priors through large-scale reinforcement learning to enable data-efficient training of vision language action models for dexterous manipulation.

NVIDIA | Isaac Loco-Manipulation Intern May 2025 – Dec 2025

Characterizes how data composition and scaling, training dynamics, and contrastive learning shape shared representations for policy transfer and generalization across human-robot embodiments.

Selected Publications

*Equal contribution. Complete publication list follows at the end.

[Spotlight] Diagnosing Compositional Generalization in Sequential Robot Tasks
Yixiao Wang, C.-E. Wu, L. Sun, P. Wang, X. Ji, B. Liang, G. Zhan, and M. Tomizuka.
CoRL, 2026.

DADP: Domain Adaptive Diffusion Policy
P. Wang*, Q. Liu*, H. Lin, Y. Li, G. Zhan, M. Tomizuka, and **Yixiao Wang**.
ICML, 2026.

VER: Vision Expert Transformer for Robot Learning via Foundation Distillation and Dynamic Routing
Yixiao Wang, M. Huo, Z. Liang, Y. Du, L. Sun, H. Lin, J. Shang, C. Peng, M. Bansal, M. Ding, and M. Tomizuka.
ICLR, 2026.

[Oral] Mean Flow Policy with Instantaneous Velocity Constraint for One-step Action Generation
G. Zhan*, L. Tao*, P. Wang, **Yixiao Wang**, Y. Chen, Y. Li, H. Li, M. Tomizuka, and S. Li.
ICLR, 2026.

[Best Paper Award in Automation] Physics-Aware Robotic Palletization with Online Masking Inference
T. Zhang, Z. Wu, Y. Chen, **Yixiao Wang**, B. Liang, S. Moura, M. Tomizuka, M. Ding, and W. Zhan.
ICRA, 2025.

Sparse Diffusion Policy: A Sparse, Reusable, and Flexible Policy for Robot Learning
Yixiao Wang*, Y. Zhang*, M. Huo*, R. Tian, X. Zhang, Y. Xie, C. Xu, P. Ji, W. Zhan, M. Ding, and M. Tomizuka.
CoRL, 2024.

Optimizing Diffusion Models for Joint Trajectory Prediction and Controllable Generation
Yixiao Wang, C. Tang, L. Sun, S. Rossi, Y. Xie, C. Peng, T. Hannagan, S. Sabatini, N. Poerio, M. Tomizuka, and W. Zhan.
ECCV, 2024.

Selected Doctoral Research

Compositional Generalization in Sequential Robot Tasks CoRL 2026 **[Spotlight]**

Derived an upper bound on the compositional generalization gap, decomposing it into marginal instruction shift, instruction compositional shift, and context-action shift, to key training data properties that govern generalization.

Domain-Adaptive Diffusion Policies ICML 2026

Domain adaptive policy that disentangles persistent physical properties, such as friction and limb length, from transient dynamics and injects the learned representations into the training and diffusion process.

Vision Expert Transformer via Foundation Distillation and Dynamic Routing ICLR 2026

Modular visual encoder that distills knowledge from multiple vision foundation models into a reusable expert library and employs lightweight dynamic routing to select task-relevant visual representations.

Sparse Diffusion Policy CoRL 2024

Unified robot model architecture based on a mixture of experts Transformer that enables specialized expert learning, selective expert activation, cross task knowledge reuse, and incremental expert expansion to support multitask learning, continual learning, and parameter efficient adaptation.

Prior-Guided Trajectory Generation and Planning ECCV 2024

Prior-guided trajectory generation and planning methods by analytically optimizing Gaussian diffusion priors for efficient sampling, introducing clean trajectory guidance for controllable generation, and integrating learned behavior priors with lightweight objective modules for efficient trajectory optimization.

Generalization Theory for Imitation Learning Preprint, 2025

An information-theoretic framework that characterizes how representation compression, model-data dependence, and training-data distributions shape generalization in imitation learning.

Technical Expertise

Robot platforms: Single- and dual-arm systems; high-DOF dexterous hands and parallel grippers.
Simulation: Large-scale reinforcement learning; Isaac Lab, MuJoCo.
Data collection & deployment: Haptic teleoperation, VR, and SpaceMouse systems.
Policy learning: Multimodal VLA models, diffusion/flow-matching policies, imitation learning.

Selected Honors

Gold Reviewer, ICML 2026
Best Paper Award in Automation, ICRA 2025
Outstanding Graduate, Shanghai Jiao Tong University 2020
Outstanding Graduate of Shanghai City 2018
Shanghai City Scholarship 2017

Professional Service

Conference reviewer ICML, ICLR, ICRA, IROS.
Journal reviewer T-RO, RA-L, TMLR.

Publications and Preprints

*Equal contribution; †corresponding author. Google Scholar.

Conference Papers

[C12] [Spotlight] Diagnosing Compositional Generalization in Sequential Robot Tasks
Yixiao Wang, C.-E. Wu, L. Sun, P. Wang, X. Ji, B. Liang, G. Zhan, and M. Tomizuka.
CoRL, 2026.

[C11] DADP: Domain Adaptive Diffusion Policy
P. Wang*, Q. Liu*, H. Lin, Y. Li, G. Zhan, M. Tomizuka, and **Yixiao Wang**.
ICML, 2026.

[C10] VER: Vision Expert Transformer for Robot Learning via Foundation Distillation and Dynamic Routing
Yixiao Wang, M. Huo, Z. Liang, Y. Du, L. Sun, H. Lin, J. Shang, C. Peng, M. Bansal, M. Ding, and M. Tomizuka.
ICLR, 2026.

[C9] Interleave-VLA: Enhancing Robot Manipulation with Image-Text Interleaved Instructions
C. Fan*, X. Jia*, Y. Sun, **Yixiao Wang**, J. Wei, Z. Gong, X. Zhao, M. Tomizuka, X. Yang, J. Yan, and M. Ding.
ICLR, 2026.

[C8] [Oral] Mean Flow Policy with Instantaneous Velocity Constraint for One-step Action Generation
G. Zhan*, L. Tao*, P. Wang, **Yixiao Wang**, Y. Chen, Y. Li, H. Li, M. Tomizuka, and S. Li.
ICLR, 2026.

[C7] DexHandDiff: Interaction-aware Diffusion Planning for Adaptive Dexterous Manipulation
Z. Liang, Y. Mu, **Yixiao Wang**, T. Chen, W. Shao, W. Zhan, M. Tomizuka, P. Luo, and M. Ding.
CVPR, 2025.

[C6] DeSiRe-GS: 4D Street Gaussians for Static-Dynamic Decomposition and Surface Reconstruction for Urban Driving Scenes
C. Peng*, C. Zhang*, **Yixiao Wang**, C. Xu, Y. Xie, W. Zheng, K. Keutzer, M. Tomizuka, and W. Zhan.
CVPR, 2025.

[C5] [Best Paper Award in Automation] Physics-Aware Robotic Palletization with Online Masking Inference
T. Zhang, Z. Wu, Y. Chen, **Yixiao Wang**, B. Liang, S. Moura, M. Tomizuka, M. Ding, and W. Zhan.
ICRA, 2025.

[C4] Adaptive Energy Regularization for Autonomous Gait Transition and Energy-Efficient Quadruped Locomotion
B. Liang*, L. Sun*, X. Zhu*, B. Zhang, Z. Xiong, **Yixiao Wang**, C. Li, K. Sreenath, and M. Tomizuka.
ICRA, 2025.

[C3] Sparse Diffusion Policy: A Sparse, Reusable, and Flexible Policy for Robot Learning
Yixiao Wang*, Y. Zhang*, M. Huo*, R. Tian, X. Zhang, Y. Xie, C. Xu, P. Ji, W. Zhan, M. Ding, and M. Tomizuka.
CoRL, 2024.

[C2] Optimizing Diffusion Models for Joint Trajectory Prediction and Controllable Generation
Yixiao Wang, C. Tang, L. Sun, S. Rossi, Y. Xie, C. Peng, T. Hannagan, S. Sabatini, N. Poerio, M. Tomizuka, and W. Zhan.
ECCV, 2024.

[C1] Joint Pedestrian Trajectory Prediction through Posterior Sampling
H. Lin, **Yixiao Wang**†, M. Huo, C. Peng, Z. Liu, and M. Tomizuka.
IROS, 2024.

Publications and Preprints (continued)

Journal Article

[J1] Singularity-Avoidant Motion Mappings for Pose-Tracking Teleoperation With Operational Jacobian Analysis
J. Zhou, B. Liang, J. Huang, H. Zhang, I. Zhang, **Yixiao Wang**, P. Abbeel, and M. Tomizuka.
IEEE/ASME Transactions on Mechatronics, 2026. Early access.

Workshop Paper

[W1] A Dexterous and Compliant Gripper With Soft Hydraulic Actuation for Microgravity Manipulation
W. Su, J. Kam, **Yixiao Wang**, and J. Zhou.
ICRA Space Robotics Workshop, 2026.

Preprints

[P8] Latent RL Priors for Data-Efficient Dexterous Manipulation
Yixiao Wang, P. Chen, M. Mukadam, S. Gayaka, M. Tomizuka, and S. Tran.
arXiv:2611.08458, 2026.

[P7] PiHand: Latent Hand-Object Interaction Priors for Physics-Based Policy Learning
N. Berndt, X. Xie, E. Duran, **Yixiao Wang**, and G. Pons-Moll.
arXiv:2610.31299, 2026.

[P6] A Deployable Four-Finger Payload for Teleoperated Free-Flying Manipulation with Astrobee
W. Su, J. Kam, Y. Nakamura, **Yixiao Wang**, J. Zhou, and M. Tomizuka.
arXiv:2609.25614, 2026.

[P5] A Reconfigurable Dual-Opposition Architecture for Single-Hand Assembly and Manipulation
W. Su, Y. Nakamura, **Yixiao Wang**, Y. Li, M. Yu, H. Qi, B. Liang, M. Tomizuka, and J. Zhou.
arXiv:2609.22871, 2026.

[P4] Multi-Camera View Scaling for Data-Efficient Robot Imitation Learning
Y. Xie, **Yixiao Wang**, S. Zhao, C.-E. Wu, M. Tomizuka, J. Xie, and H.-S. Fang.
arXiv:2604.00557, 2026.

[P3] Generalization Capability for Imitation Learning
Yixiao Wang.
arXiv:2504.18538, 2025.

[P2] Imagined Potential Games: A Framework for Simulating, Learning and Evaluating Interactive Behaviors
L. Sun, **Yixiao Wang**, P.-Y. Hung, C. Wang, X. Zhang, Z. Xu, and M. Tomizuka.
arXiv:2411.03669, 2024.

[P1] Composition Vision-Language Understanding via Segment and Depth Anything Model
M. Huo, P. Ji, H. Lin, J. Liu, **Yixiao Wang**, and Y. Chen.
arXiv:2406.18591, 2024.